

DOES 1C3-DIT/DENG11G1 – Bioinformatics

RNN LSTM GRU

Assignment October 4, 2026

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1. RNN Layer

Architecture Type A: One – to – One

Model:

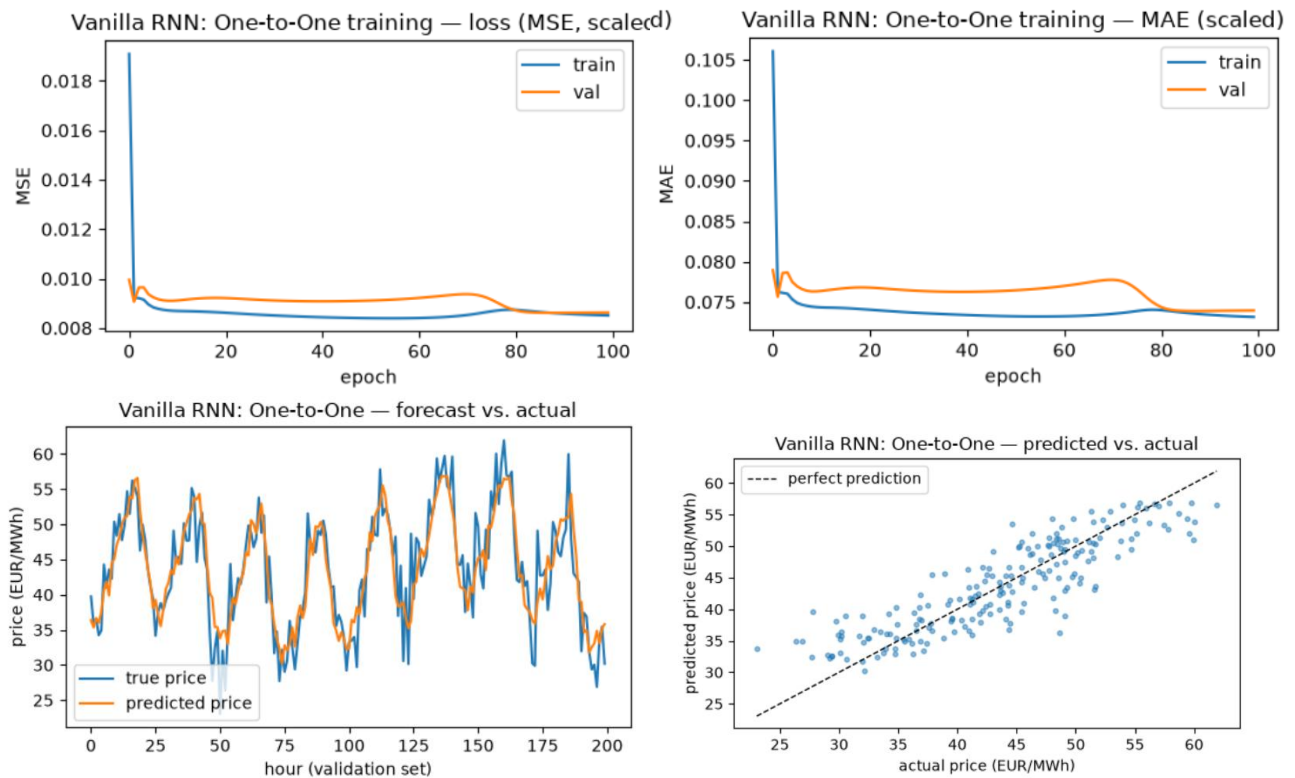
Layer (type)	Output Shape	Param #
simple_rnn (SimpleRNN)	(None, 1, 32)	1,216
simple_rnn_1 (SimpleRNN)	(None, 1, 32)	2,080
simple_rnn_2 (SimpleRNN)	(None, 32)	2,080
dense (Dense)	(None, 1)	33

Total params: 5,409 (21.13 KB)

Trainable params: 5,409 (21.13 KB)

Non-trainable params: 0 (0.00 B)

one_to_one: ran 100/100 epochs final val_loss=0.00863 val_mae=0.07394 params=5,409



Architecture Type B: Many – to – One

Model:

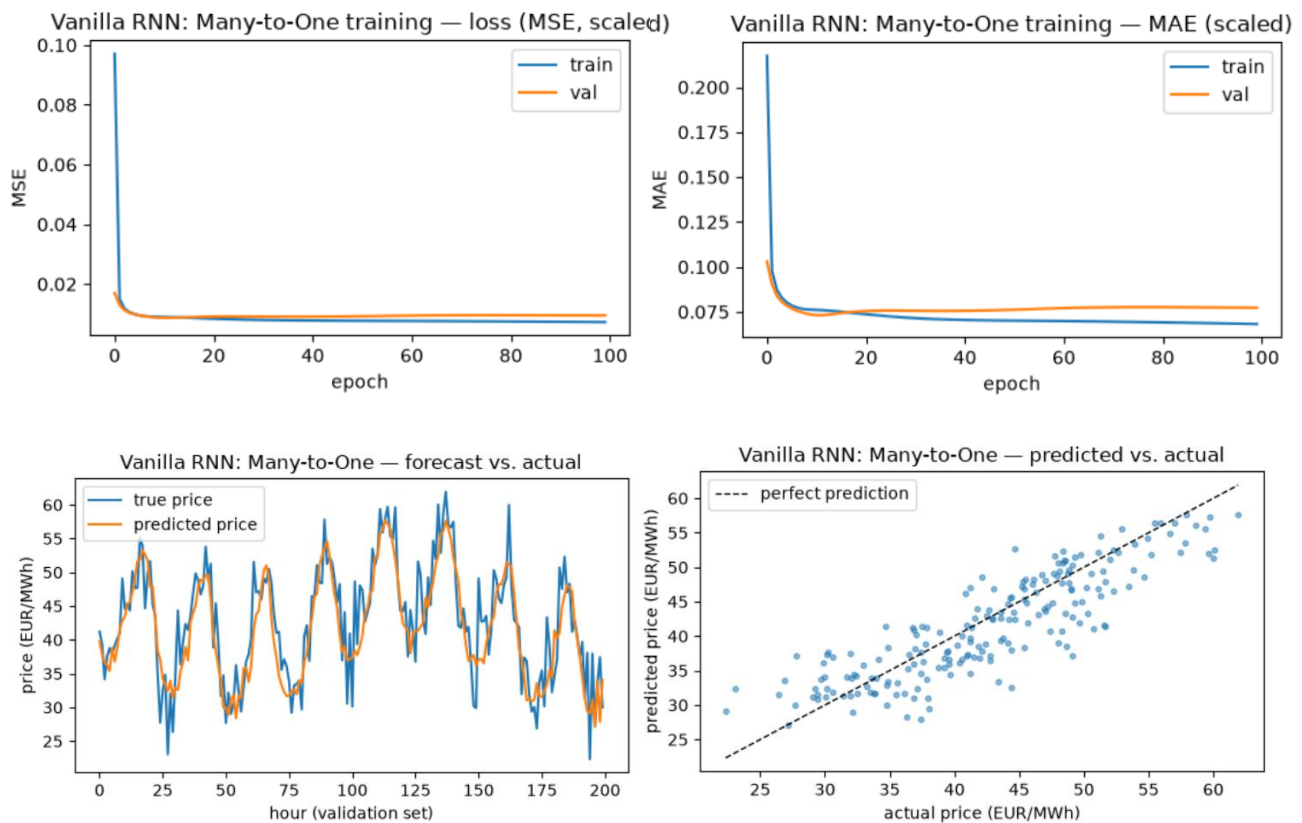
Layer (type)	Output Shape	Param #
simple_rnn_5 (SimpleRNN)	(None, 24, 32)	1,216
simple_rnn_6 (SimpleRNN)	(None, 32)	2,080
dense_2 (Dense)	(None, 1)	33

Total params: 3,329 (13.00 KB)

Trainable params: 3,329 (13.00 KB)

Non-trainable params: 0 (0.00 B)

many_to_one: ran 100/100 epochs final val_loss=0.00951 val_mae=0.07725 params=3,329



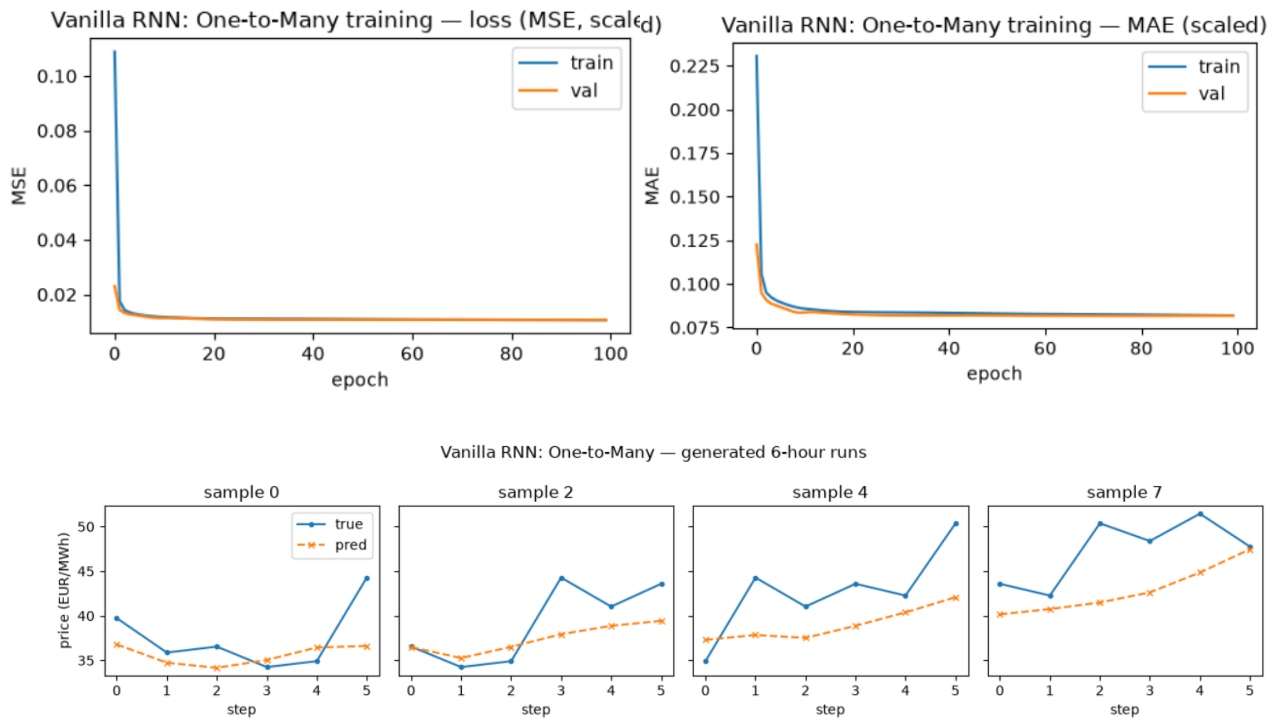
Architecture Type C: One – to – Many

Model:

Layer (type)	Output Shape	Param #
simple_rnn_5 (SimpleRNN)	(None, 1, 32)	1,216
simple_rnn_6 (SimpleRNN)	(None, 32)	2,080
repeat_vector (RepeatVector)	(None, 6, 32)	0
simple_rnn_7 (SimpleRNN)	(None, 6, 32)	2,080
simple_rnn_8 (SimpleRNN)	(None, 6, 32)	2,080
time_distributed (TimeDistributed)	(None, 6, 1)	33

Total params: 7,489 (29.25 KB)
Trainable params: 7,489 (29.25 KB)
Non-trainable params: 0 (0.00 B)

one_to_many: ran 100/100 epochs final val_loss=0.01074 val_mae=0.08174 params=7,489

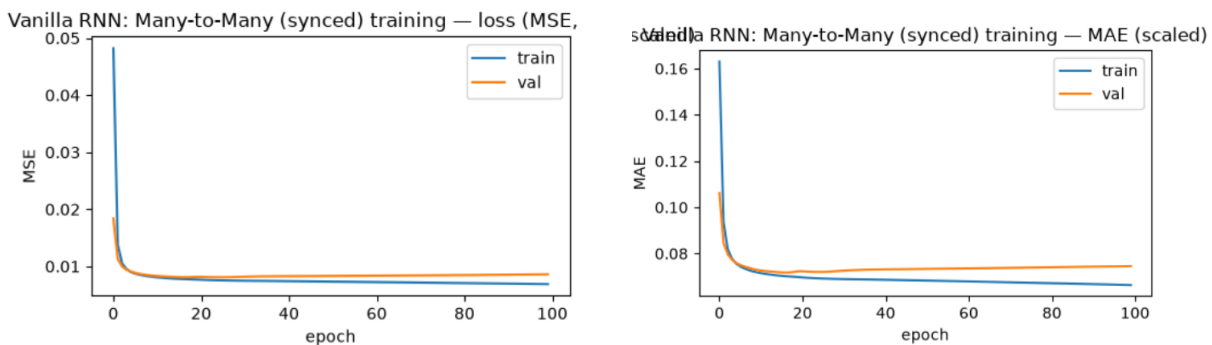


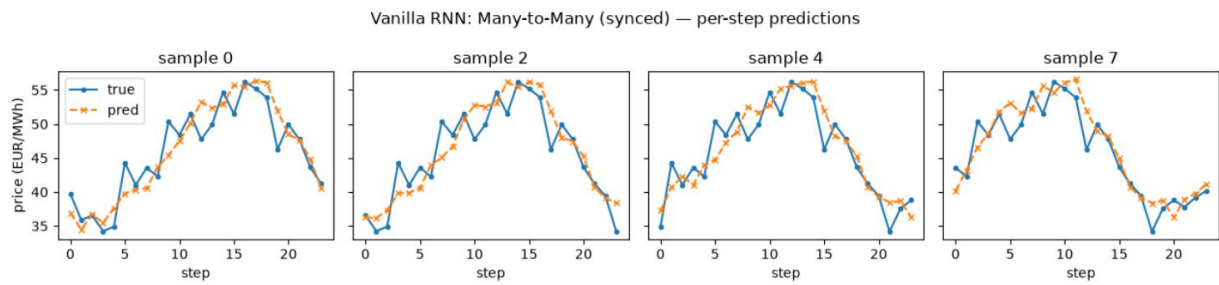
Architecture Type D: Many – to – Many (Synced)

Model:

Layer (type)	Output Shape	Param #
simple_rnn_9 (SimpleRNN)	(None, 24, 32)	1,216
simple_rnn_10 (SimpleRNN)	(None, 24, 32)	2,080
time_distributed_1 (TimeDistributed)	(None, 24, 1)	33

Total params: 3,329 (13.00 KB)
Trainable params: 3,329 (13.00 KB)
Non-trainable params: 0 (0.00 B)





Architecture Type E: Many – to – Many (Encoder–Decoder / Seq2Seq)

Model:

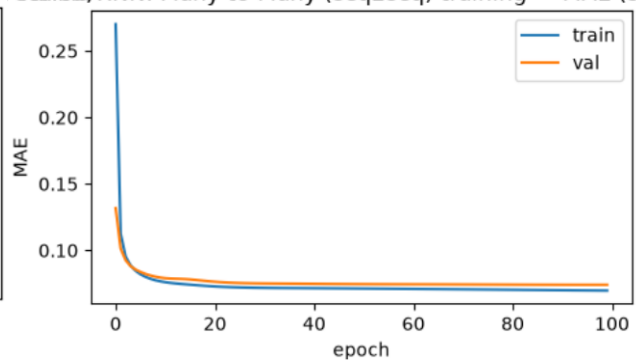
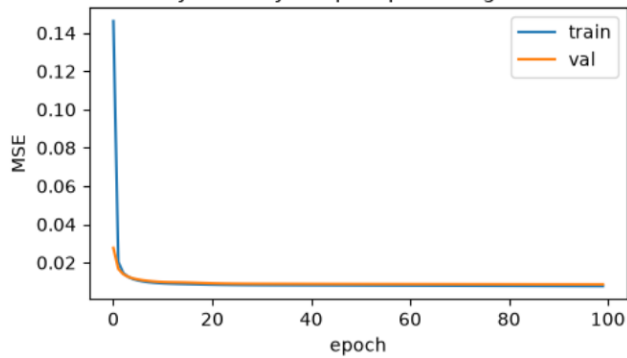
Layer (type)	Output Shape	Param #
simple_rnn_11 (SimpleRNN)	(None, 24, 32)	1,216
simple_rnn_12 (SimpleRNN)	(None, 32)	2,080
repeat_vector_1 (RepeatVector)	(None, 6, 32)	0
simple_rnn_13 (SimpleRNN)	(None, 6, 32)	2,080
simple_rnn_14 (SimpleRNN)	(None, 6, 32)	2,080
time_distributed_2 (TimeDistributed)	(None, 6, 1)	33

Total params: 7,489 (29.25 KB)

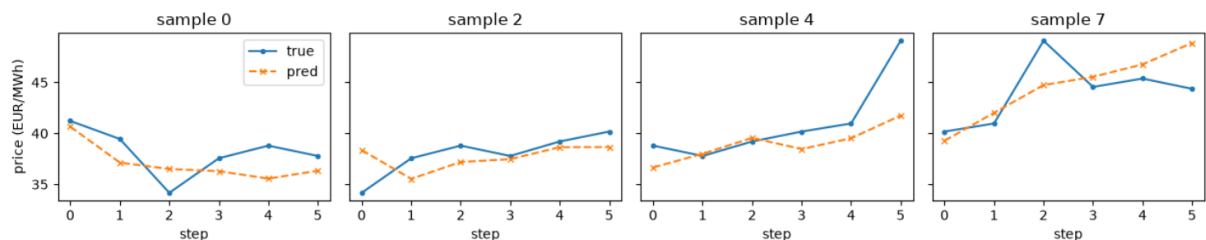
Trainable params: 7,489 (29.25 KB)

Non-trainable params: 0 (0.00 B)

Vanilla RNN: Many-to-Many (seq2seq) training — loss (MSE) (Vanilla RNN: Many-to-Many (seq2seq) training — MAE (sc



Vanilla RNN: Many-to-Many (seq2seq) — 24h -> 6h forecast



RNN: Performance Metrics

	MAE	MSE	RMSE	R ² Score
Architecture Type				
One-to-One	3.4813	18.2107	4.2674	0.7272
Many-to-One	3.3094	17.4592	4.1784	0.7368
One-to-Many	3.7003	21.8275	4.6720	0.6744
Many-to-Many (Synced)	3.3011	16.8317	4.1027	0.7490
Many-to-Many (Seq2Seq)	3.2358	16.4720	4.0586	0.7491

The performance metrics (Mean Absolute Error (MAE), Mean Squared Error (MSE), Root Mean Squared Error (RMSE), and R² Score) show clear differences in performance depending on how inputs and outputs are structured across sequential time steps. Lower values for MAE, MSE, and RMSE indicate smaller prediction errors, while a higher R² Score indicates a better fit and higher accuracy in explaining variance.

The One-to-Many architecture yielded the weakest performance among all models. It generated the highest errors across all metrics, with an MAE of 3.7003, an MSE of 21.8275, an RMSE of 4.6720, and the lowest R² score of 0.6744. Because a One-to-Many setup relies on a single input vector to generate multiple outputs, the network receives limited input context over time, making it harder to maintain accurate long-term sequential predictions.

The One-to-One architecture showed baseline improvement over One-to-Many, recording an MAE of 3.4813, MSE of 18.2107, RMSE of 4.2674, and an R² Score of 0.7272. Although it processes single inputs to produce single outputs without full sequential context, adding stacked RNN layers still allows it to capture basic patterns reasonably well.

The Many-to-One architecture demonstrated significant improvements over the single-input models, producing an MAE of 3.3094, an MSE of 17.4592, an RMSE of 4.1784, and an R² Score of 0.7368. By consuming multiple time steps before making a single prediction, the model benefits from a richer temporal context, significantly reducing error rates.

Both Many-to-Many architectures outperformed the previous models. The Many-to-Many (Synced) model achieved an MAE of 3.3011, MSE of 16.8317, RMSE of 4.1027, and an R² Score of 0.7490. It benefits from updating predictions at each time step while continuously taking sequence inputs, ensuring feedback loops remain closely aligned with real-time data.

The Many-to-Many (Seq2Seq/Encoder–Decoder) architecture performs best overall.

2. LSTM Layer

Architecture Type A: One – to – One

Model:

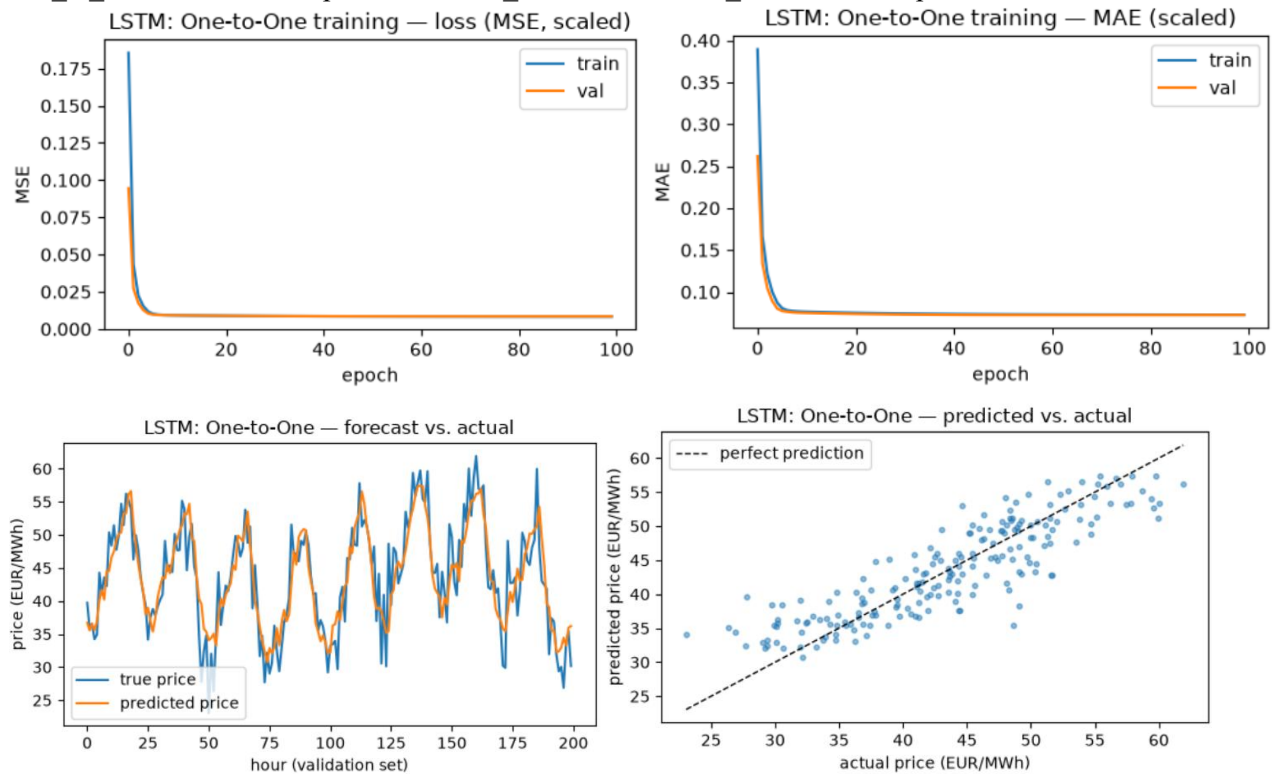
Layer (type)	Output Shape	Param #
lstm (LSTM)	(None, 1, 32)	4,864
lstm_1 (LSTM)	(None, 32)	8,320
dense (Dense)	(None, 1)	33

Total params: 13,217 (51.63 KB)

Trainable params: 13,217 (51.63 KB)

Non-trainable params: 0 (0.00 B)

one_to_one: ran 100/100 epochs final val_loss=0.00849 val_mae=0.07265 params=13,217



Architecture Type B: Many – to – One

Model:

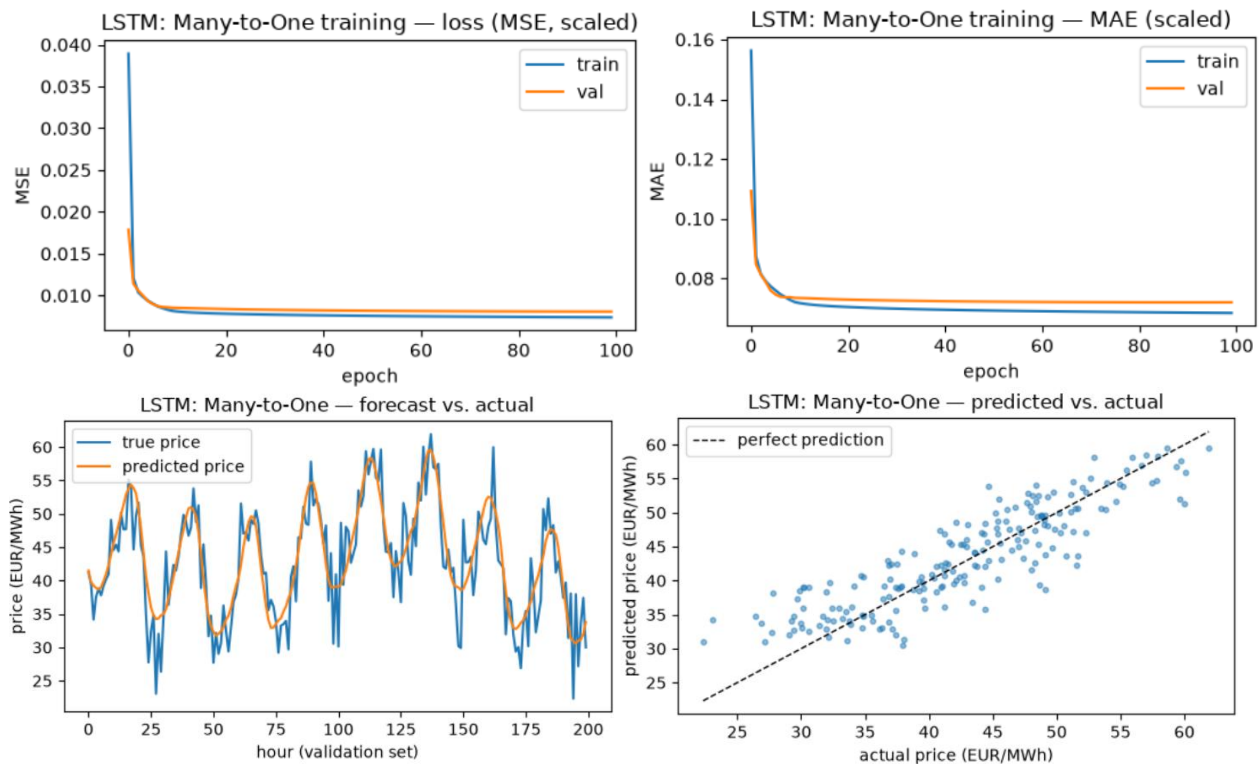
Layer (type)	Output Shape	Param #
lstm_2 (LSTM)	(None, 24, 32)	4,864
lstm_3 (LSTM)	(None, 32)	8,320
dense_1 (Dense)	(None, 1)	33

Total params: 13,217 (51.63 KB)

Trainable params: 13,217 (51.63 KB)

Non-trainable params: 0 (0.00 B)

many_to_one: ran 100/100 epochs final val_loss=0.00807 val_mae=0.07204 params=13,217



Architecture Type C: One – to – Many

Model:

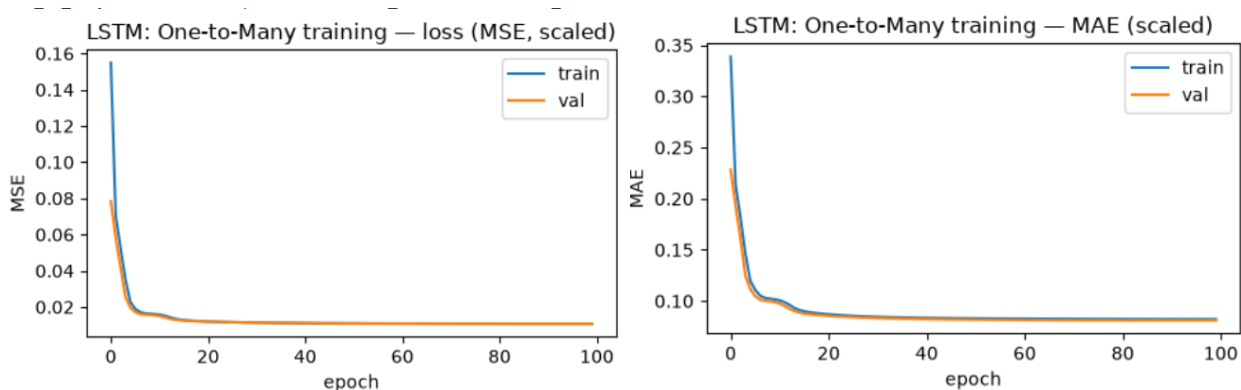
Layer (type)	Output Shape	Param #
lstm_4 (LSTM)	(None, 1, 32)	4,864
lstm_5 (LSTM)	(None, 32)	8,320
repeat_vector (RepeatVector)	(None, 6, 32)	0
lstm_6 (LSTM)	(None, 6, 32)	8,320
lstm_7 (LSTM)	(None, 6, 32)	8,320
time_distributed (TimeDistributed)	(None, 6, 1)	33

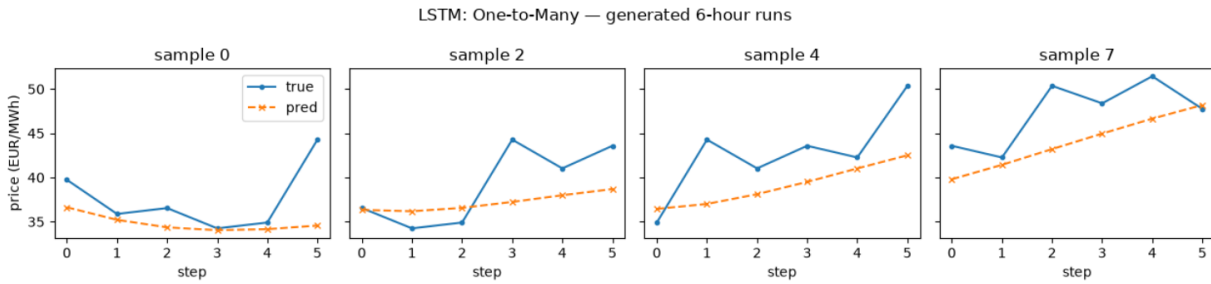
Total params: 29,857 (116.63 KB)

Trainable params: 29,857 (116.63 KB)

Non-trainable params: 0 (0.00 B)

one_to_many: ran 100/100 epochs final val_loss=0.01064 val_mae=0.08104 params=29,857





Architecture Type D: Many – to – Many (Synced)

Model:

Layer (type)	Output Shape	Param #
lstm_8 (LSTM)	(None, 24, 32)	4,864
lstm_9 (LSTM)	(None, 24, 32)	8,320
time_distributed_1 (TimeDistributed)	(None, 24, 1)	33

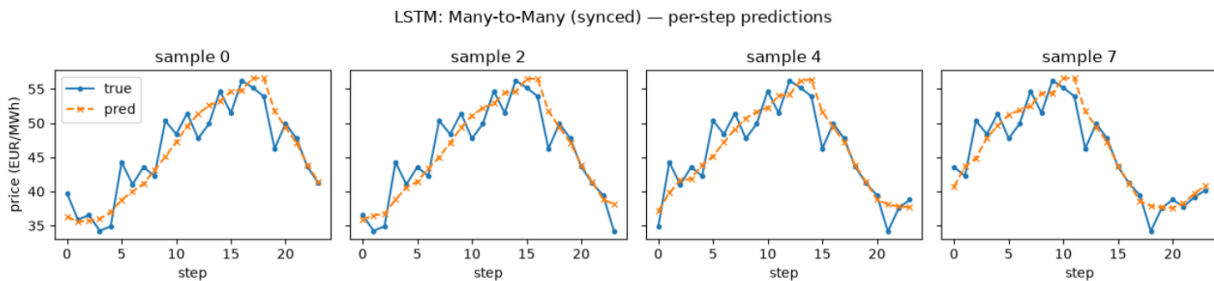
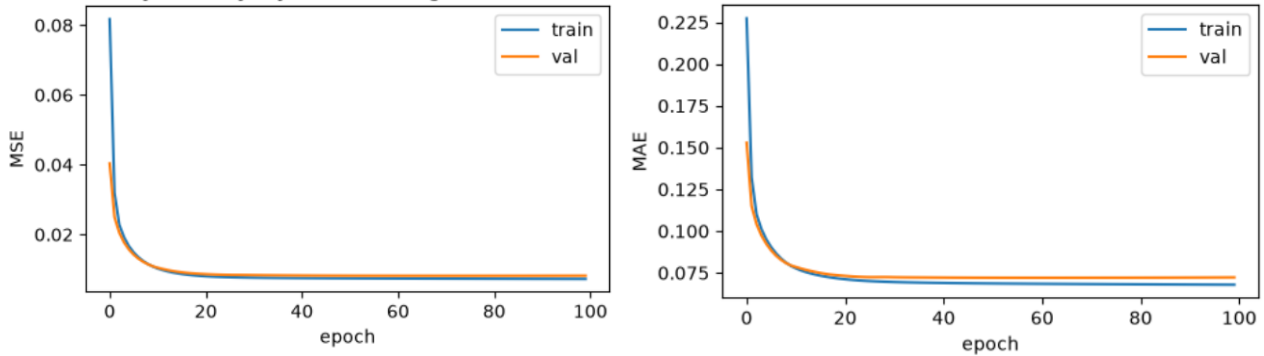
Total params: 13,217 (51.63 KB)

Trainable params: 13,217 (51.63 KB)

Non-trainable params: 0 (0.00 B)

many_to_many_synced: ran 100/100 epochs final val_loss=0.00820 val_mae=0.07230 params=13,217

LSTM: Many-to-Many (synced) training — loss (MSE, scaled) LSTM: Many-to-Many (synced) training — MAE (scaled)

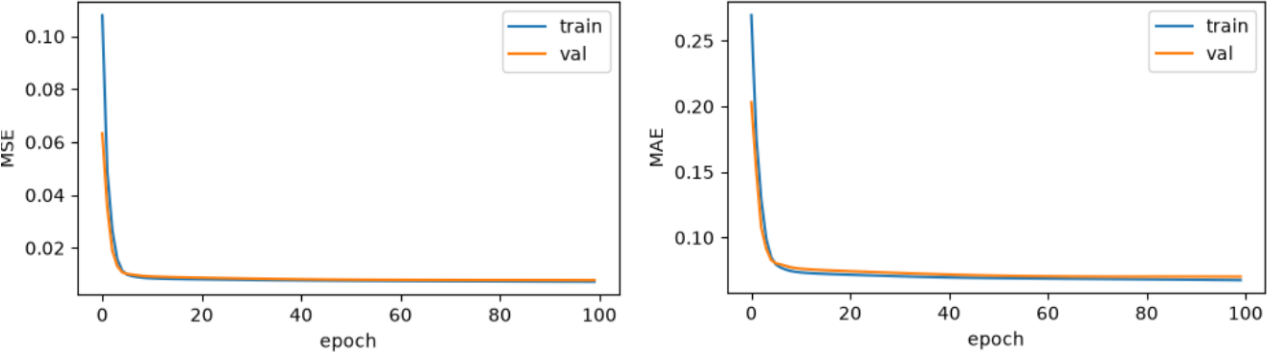


Architecture Type E: Many – to – Many ()
Model:

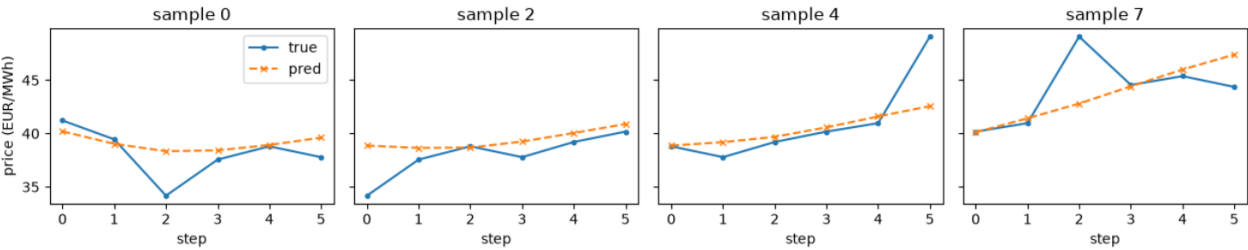
Layer (type)	Output Shape	Param #
lstm_10 (LSTM)	(None, 24, 32)	4,864
lstm_11 (LSTM)	(None, 32)	8,320
repeat_vector_1 (RepeatVector)	(None, 6, 32)	0
lstm_12 (LSTM)	(None, 6, 32)	8,320
lstm_13 (LSTM)	(None, 6, 32)	8,320
time_distributed_2 (TimeDistributed)	(None, 6, 1)	33

Total params: 29,857 (116.63 KB)
Trainable params: 29,857 (116.63 KB)
Non-trainable params: 0 (0.00 B)

LSTM: Many-to-Many (seq2seq) training — loss (MSE, scaled)LSTM: Many-to-Many (seq2seq) training — MAE (scaled)



LSTM: Many-to-Many (seq2seq) — 24h -> 6h forecast



LSTM: Performance Metrics

	MAE	MSE	RMSE	R ² Score
Architecture Type				
One-to-One	3.2853	17.1184	4.1374	0.7436
Many-to-One	3.1748	15.3319	3.9156	0.7689
One-to-Many	3.7001	21.9264	4.6826	0.6729
Many-to-Many (Synced)	3.2100	15.8718	3.9839	0.7633
Many-to-Many (Seq2Seq)	3.1249	15.2741	3.9082	0.7674

The One-to-Many architecture remains the weakest performer overall, showing an MAE of 3.7001, MSE of 21.9264, RMSE of 4.6826, and an R² Score of 0.6729. Because the model relies on a single initial input vector to predict a sequence of outputs, even adding LSTM cells cannot compensate for the lack of continuous input features over time.

The One-to-One model demonstrates noticeable improvements with LSTM, recording an MAE of 3.2853, MSE of 17.1184, RMSE of 4.1374, and an R² Score of 0.7436. The LSTM's internal memory state allows it to better map single inputs to target outputs compared to standard RNNs.

The Many-to-Many (Synced) architecture shows solid performance across all metrics with an MAE of 3.2100, MSE of 15.8718, RMSE of 3.9839, and an R² Score of 0.7633. Synchronized mapping allows the LSTM cell states to maintain temporal context while making predictions at every single time step.

The Many-to-One and Many-to-Many (Seq2Seq) architectures stand out as the top two performers overall. Many-to-One achieves the highest explanatory power with an R² Score of 0.7689, alongside strong error metrics (MAE: 3.1748, MSE: 15.3319, RMSE: 3.9156). It excels at aggregating multi-step historical sequence inputs into a single target prediction.

The Many-to-Many (Seq2Seq / Encoder–Decoder) architecture achieves the lowest overall prediction error, while Many-to-One yields the highest R² score.

3. GRU Layer

Architecture Type A: One – to – One

Model:

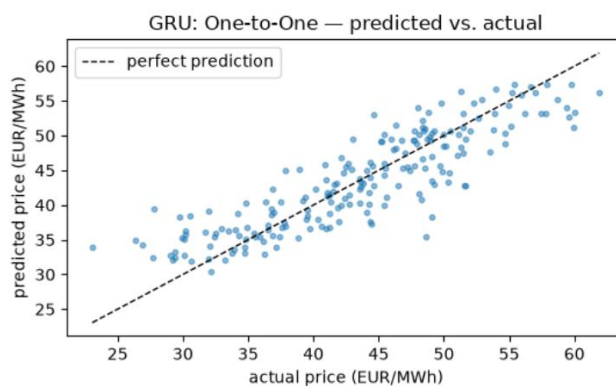
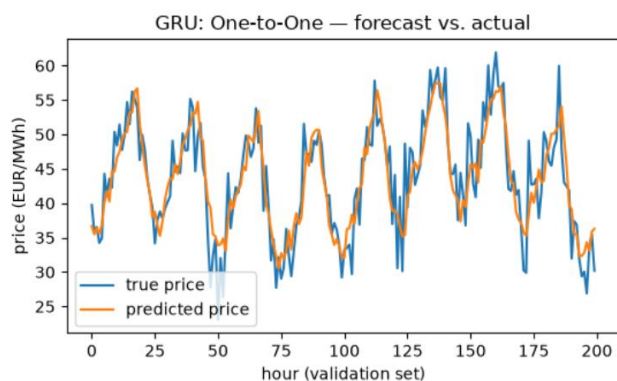
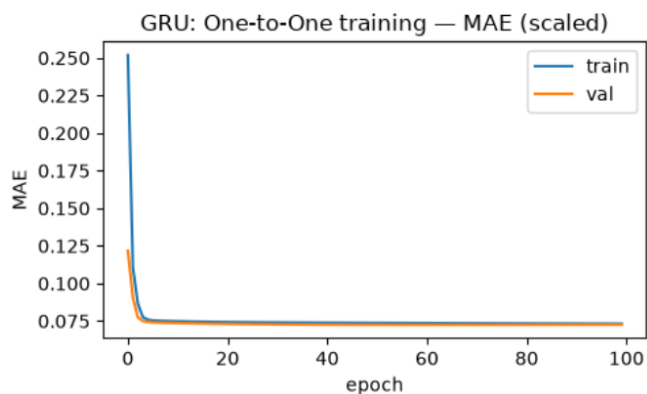
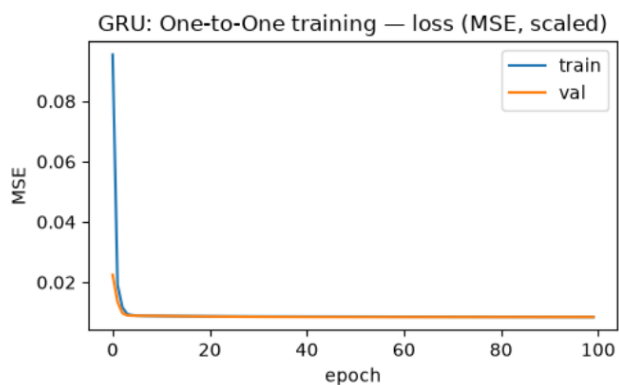
Layer (type)	Output Shape	Param #
gru (GRU)	(None, 1, 32)	3,744
gru_1 (GRU)	(None, 32)	6,336
dense (Dense)	(None, 1)	33

Total params: 10,113 (39.50 KB)

Trainable params: 10,113 (39.50 KB)

Non-trainable params: 0 (0.00 B)

one_to_one: ran 100/100 epochs final val_loss=0.00848 val_mae=0.07264 params=10,113



Architecture Type B: Many – to – One

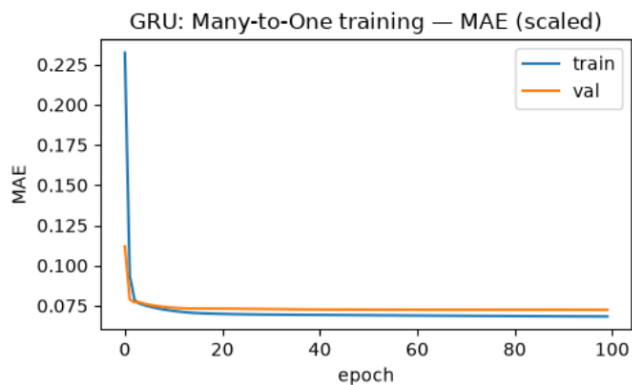
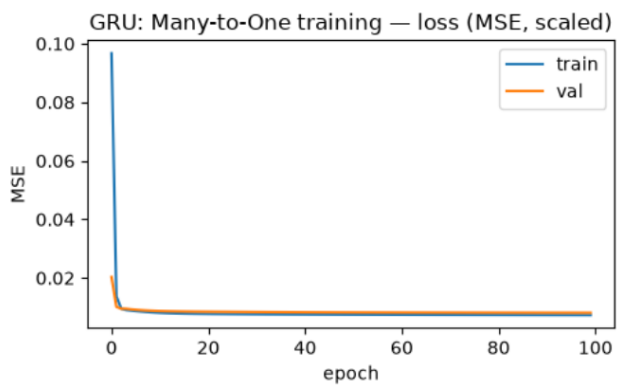
Model:

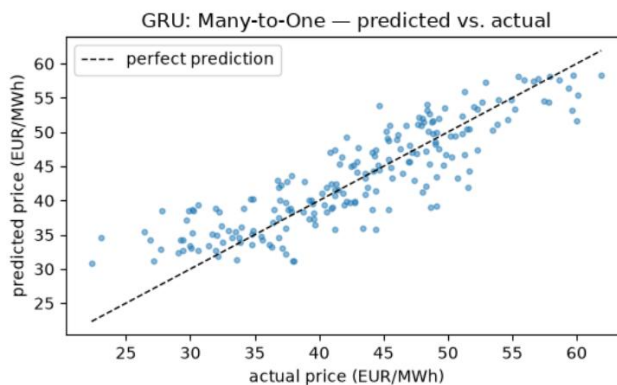
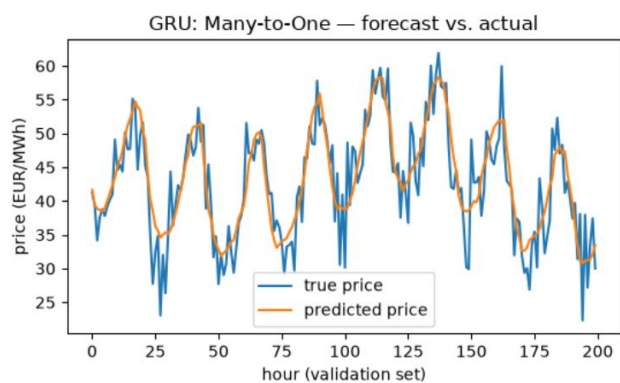
Layer (type)	Output Shape	Param #
gru_2 (GRU)	(None, 24, 32)	3,744
gru_3 (GRU)	(None, 32)	6,336
dense_1 (Dense)	(None, 1)	33

Total params: 10,113 (39.50 KB)

Trainable params: 10,113 (39.50 KB)

Non-trainable params: 0 (0.00 B)





Architecture Type C: One – to – Many

Model:

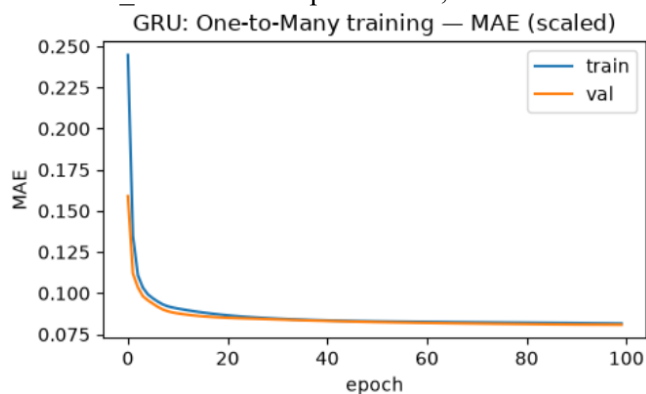
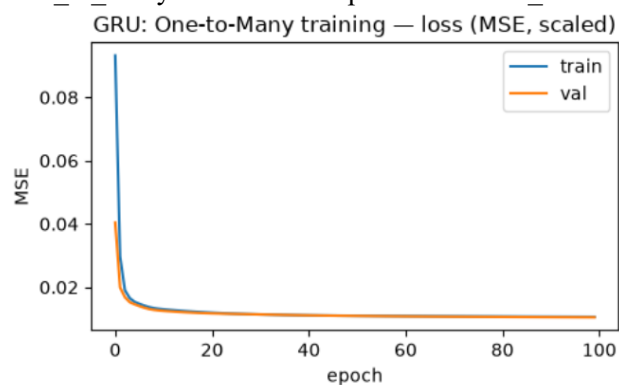
Layer (type)	Output Shape	Param #
gru_4 (GRU)	(None, 1, 32)	3,744
gru_5 (GRU)	(None, 32)	6,336
repeat_vector (RepeatVector)	(None, 6, 32)	0
gru_6 (GRU)	(None, 6, 32)	6,336
gru_7 (GRU)	(None, 6, 32)	6,336
time_distributed (TimeDistributed)	(None, 6, 1)	33

Total params: 22,785 (89.00 KB)

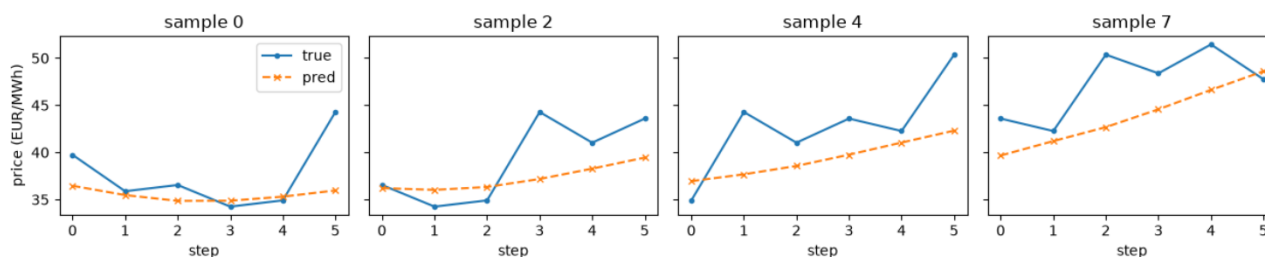
Trainable params: 22,785 (89.00 KB)

Non-trainable params: 0 (0.00 B)

one_to_many: ran 100/100 epochs final val_loss=0.01048 val_mae=0.08087 params=22,785



GRU: One-to-Many — generated 6-hour runs



Architecture Type D: Many – to – Many (Synced)

Model:

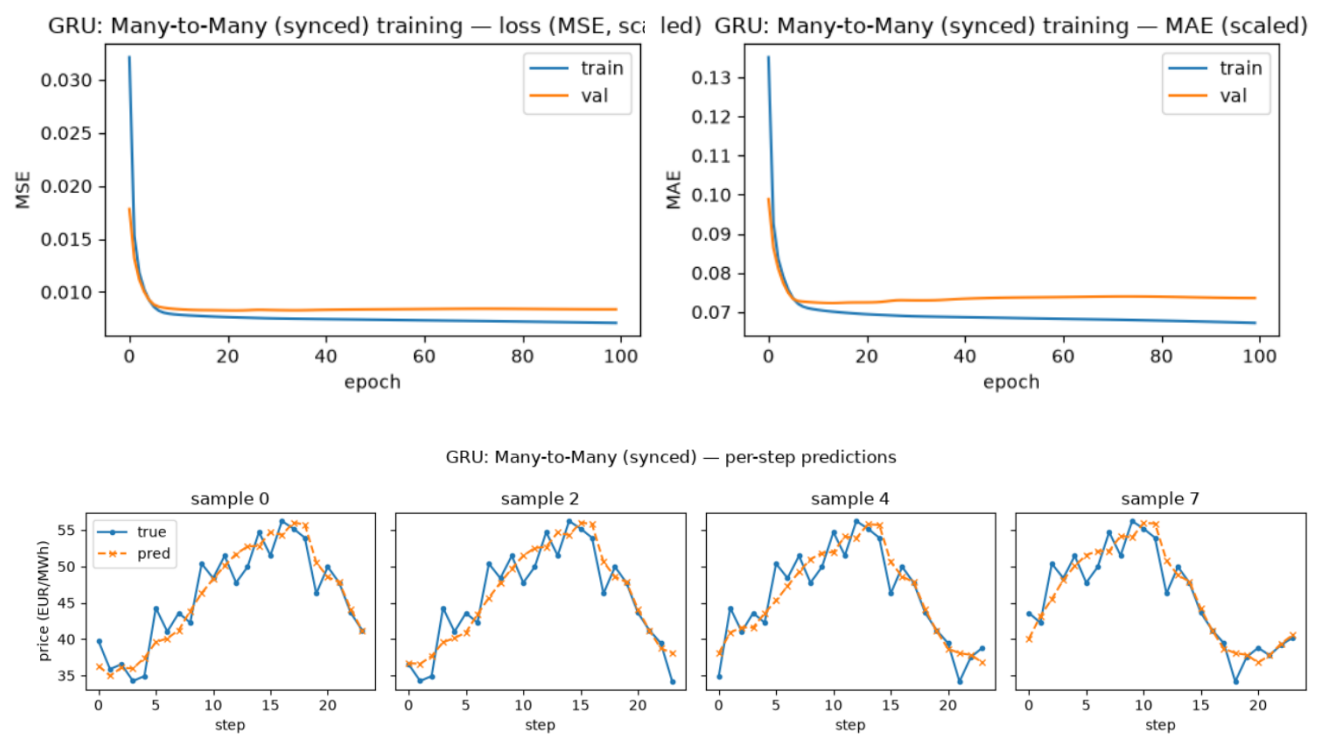
Layer (type)	Output Shape	Param #
gru_8 (GRU)	(None, 24, 32)	3,744
gru_9 (GRU)	(None, 24, 32)	6,336
time_distributed_1 (TimeDistributed)	(None, 24, 1)	33

Total params: 10,113 (39.50 KB)

Trainable params: 10,113 (39.50 KB)

Non-trainable params: 0 (0.00 B)

many_to_many_synced: ran 100/100 epochs final val_loss=0.00836 val_mae=0.07354 params=10,113



Architecture Type E: Many – to – Many ()

Model:

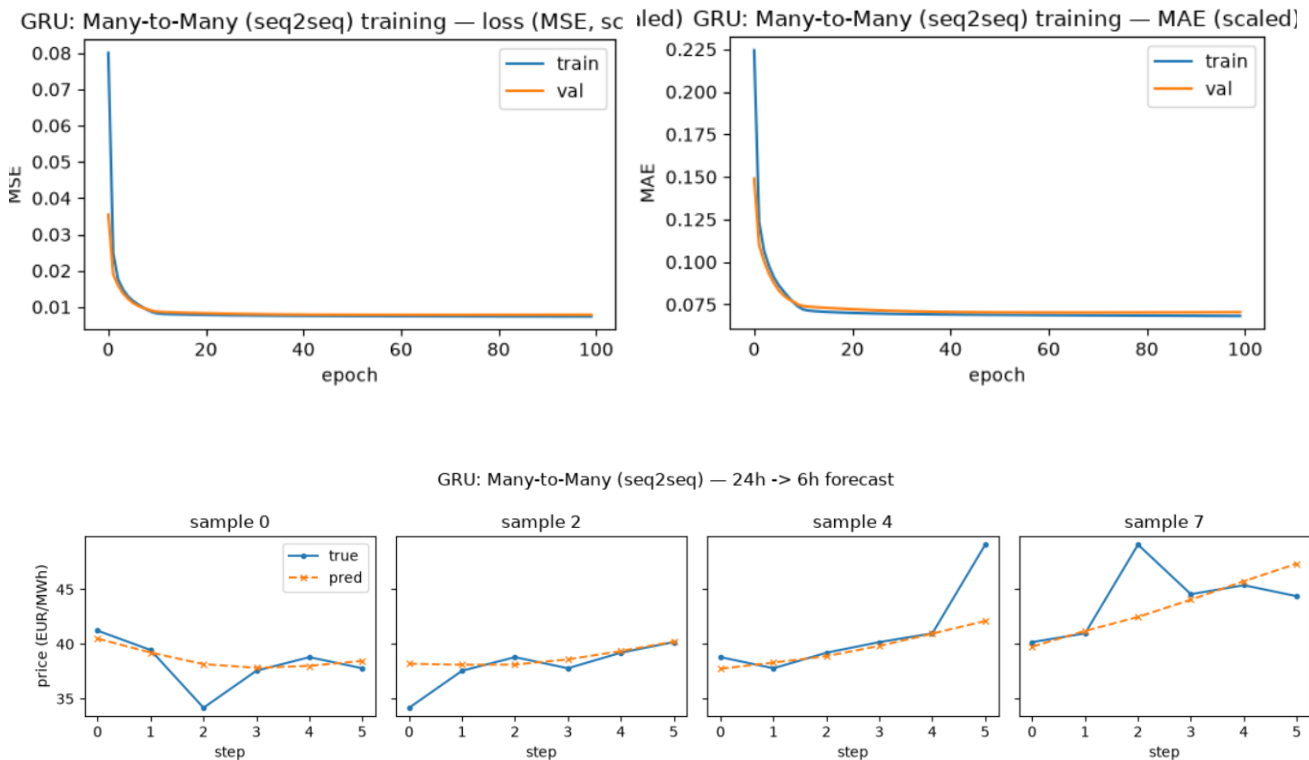
Layer (type)	Output Shape	Param #
gru_10 (GRU)	(None, 24, 32)	3,744
gru_11 (GRU)	(None, 32)	6,336
repeat_vector_1 (RepeatVector)	(None, 6, 32)	0
gru_12 (GRU)	(None, 6, 32)	6,336
gru_13 (GRU)	(None, 6, 32)	6,336
time_distributed_2 (TimeDistributed)	(None, 6, 1)	33

Total params: 22,785 (89.00 KB)

Trainable params: 22,785 (89.00 KB)

Non-trainable params: 0 (0.00 B)

many_to_many_seq2seq: ran 100/100 epochs final val_loss=0.00785 val_mae=0.07053 params=22,785



GRU: Performance Metrics

	MAE	MSE	RMSE	R ² Score
Architecture Type				
One-to-One	3.2882	17.1510	4.1414	0.7431
Many-to-One	3.1866	15.4045	3.9249	0.7678
One-to-Many	3.6499	21.1992	4.6043	0.6838
Many-to-Many (Synced)	3.2718	16.3908	4.0486	0.7555
Many-to-Many (Seq2Seq)	3.0739	14.7885	3.8456	0.7748

The One-to-Many architecture again registers the highest error rates and lowest accuracy among all models, with an MAE of 3.6499, MSE of 21.1992, RMSE of 4.6043, and an R² Score of 0.6838. Although GRU gates slightly improve its performance compared to standard RNN and LSTM versions, receiving only a single input vector limits its ability to reliably forecast an extended output sequence.

The One-to-One architecture produces stable baseline results, recording an MAE of 3.2882, MSE of 17.1510, RMSE of 4.1414, and an R² Score of 0.7431. This shows that GRUs can map individual input features to output targets effectively, though without multi-step sequence context.

The Many-to-Many (Synced) model delivers moderate performance across all evaluation criteria, achieving an MAE of 3.2718, an MSE of 16.3908, an RMSE of 4.0486, and an R^2 Score of 0.7555. While synchronized sequence-to-sequence mapping benefits from step-by-step updates, it performs slightly behind models that process input sequences in a dedicated encoding phase.

The Many-to-One model demonstrates strong predictive capability, achieving an MAE of 3.1866, MSE of 15.4045, RMSE of 3.9249, and an R^2 Score of 0.7678. Processing full historical sequences to generate a single target output enables GRU layers to effectively extract essential temporal context.

The Many-to-Many (Seq2Seq / Encoder–Decoder) architecture is the best-performing model across all evaluated metrics.

Comparative Performance Breakdown

Standard Recurrent Neural Networks (RNNs) produced the lowest performance overall across all metrics. Traditional RNNs lack specialized memory-control mechanisms, leaving them vulnerable to vanishing and exploding gradients. As a result, standard RNNs struggle to preserve long-term context across time steps, yielding higher prediction errors and lower R^2 scores compared to gated variants.

Long Short-Term Memory (LSTM) networks introduced significant improvements over standard RNNs. By using three separate gates (input, forget, and output) alongside a dedicated cell state, LSTMs effectively regulate how information is stored, updated, and discarded over sequence steps. This design enabled notable reductions in MAE, MSE, and RMSE, while pushing R^2 scores higher (peaking at 0.7689 in the Many-to-One layout and 0.7674 in Seq2Seq).

Gated Recurrent Units (GRUs) achieved the best overall performance across the benchmarked architectures. By streamlining the gating structure into two gates (reset and update) and merging the cell state with the hidden state, GRUs capture long-term dependencies with less parameter overhead and smoother gradient flow. This allowed the GRU models to achieve the overall best prediction metrics across all experiments—most notably in the Many-to-Many (Seq2Seq) configuration, which recorded the absolute lowest error metrics (MAE: 3.0739, MSE: 14.7885, RMSE: 3.8456) and the highest overall fit (R^2 : 0.7748).

The GRU-based Many-to-Many (Seq2Seq) architecture demonstrated superior performance across all evaluation parameters. By combining an efficient gating mechanism with an Encoder–Decoder framework, it effectively compresses and reconstructs temporal information while mitigating information loss, rendering it the most precise and robust model among the tested configurations.

Advanced Challenge:

Add **one CNN layer** before the RNN layer and run the model for **100 epochs** using the following settings:

- Number of filters: **32**
- Filter/kernel size: **3**

Performance Parameters

RNN						CNN - RNN					
		MAE	MSE	RMSE	R ² Score			MAE	MSE	RMSE	R ² Score
Architecture Type						Architecture Type					
One-to-One		3.4813	18.2107	4.2674	0.7272	One-to-One		3.2970	17.2919	4.1583	0.7410
Many-to-One		3.3094	17.4592	4.1784	0.7368	Many-to-One		3.7338	23.2440	4.8212	0.6496
One-to-Many		3.7003	21.8275	4.6720	0.6744	One-to-Many		3.6685	21.3374	4.6192	0.6817
Many-to-Many (Synced)		3.3011	16.8317	4.1027	0.7490	Many-to-Many (Synced)		0.2944	0.7419	0.8613	0.9889
Many-to-Many (Seq2Seq)		3.2358	16.4720	4.0586	0.7491	Many-to-Many (Seq2Seq)		3.2352	16.3213	4.0400	0.7514

When a Convolutional Neural Network (CNN) layer was added before the RNN layers, the overall performance dynamics shifted significantly across architectures. The Many-to-Many (Synced) architecture achieved the absolute best overall performance across both experiments by a wide margin. Its metrics improved dramatically to an MAE of 0.2944, an MSE of 0.7419, an RMSE of 0.8613, and an outstanding R^2 score of 0.9889. The Many-to-Many (Seq2Seq) model saw minor improvements, yielding an MAE of 3.2352, an RMSE of 4.0400, and an R^2 score of 0.7514. The One-to-One model also benefited slightly from the CNN layer, improving its R^2 score from 0.7272 to 0.7410.

However, adding a CNN layer did not improve all architectures. The Many-to-One architecture experienced significant degradation in performance, with its R^2 score dropping from 0.7368 down to 0.6496, and its RMSE increasing from 4.1784 to 4.8212. The One-to-Many model showed slight improvements in error metrics (MAE of 3.6685, R^2 of 0.6817), but still remained among the lower-performing configurations.

Across all tested configurations, the Many-to-Many (Synced) architecture with a CNN-RNN layer combination is the best-performing.

The dramatic increase in performance for the Many-to-Many (Synced) architecture when paired with a CNN layer stems from the complementary strengths of CNNs and RNNs. The initial CNN layer excels at spatial feature extraction, localized pattern recognition, and noise filtering directly from raw temporal input sequences. When synchronized step-by-step with the subsequent RNN layer, the model can leverage high-level, feature-extracted temporal representations at every timestep, resulting in near-perfect predictive accuracy ($R^2 = 0.9889$) and a massive reduction in prediction error.